IPT 2021 · MOUSETRAP CHAIN REACTIONS · NOTEBOOKS 5–6

From mousetraps to populations and fission

One branching process on a lattice, read three ways — and what mean-field theory gets right and wrong in each.

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One model, three readings

Mousetrap	Population (NB 5)	Fission (NB 6)
armed trap	susceptible / empty patch	fissile nucleus
firing trap	infected, reproducing	fission this generation
spent trap	recovered / dead	fission products
m attempts	offspring attempts	ν neutrons, $\bar{\nu} = 2.41$
f_k, p_long	dispersal kernel	flight-length distribution
ρκ	R_0	$k_\infty = \rho\, \bar{\nu}\, p_f$

The threshold is the same in all three: **ρκ = 1**, $R_0 = 1$, $k = 1$.

One new parameter: re-arming

A spent trap is re-armed with probability **p_regen** per generation. The cascade no longer consumes itself: the lattice becomes a spatial stochastic **SIRS** model.

p_regen = 0 recovers the original model; F = 1 − e^(−ρκF) is the Kermack–McKendrick final size.

Mean-field map

new = s (1 − e^(−κ i))

s′ = s − new + p_regen r

i′ = new

r′ = r + i − p_regen r

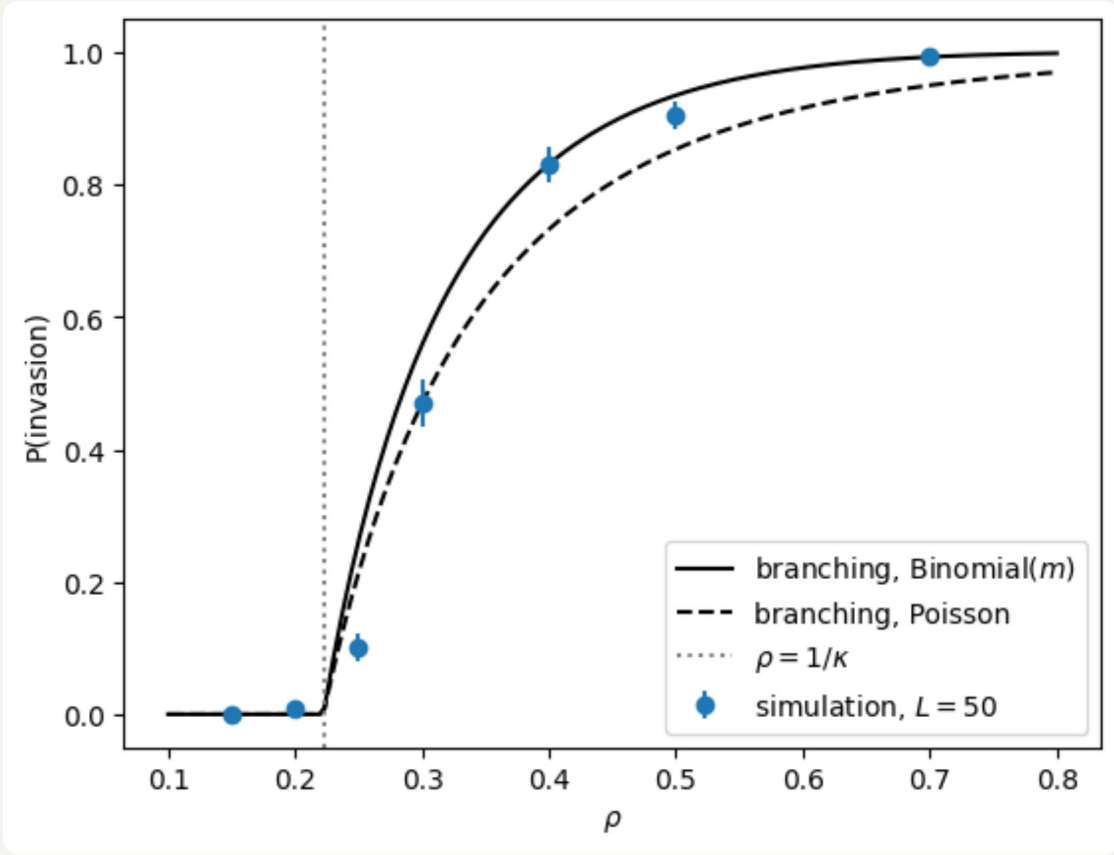
s + i + r = ρ (armed, firing, spent)

Invasion: R0 > 1 is necessary, not sufficient

Threshold confirmed at $\rho = 1/\kappa = 0.222$.

Binomial(m) beats Poisson for $\rho \geq 0.4$: lower offspring variance, fewer unlucky early extinctions.

Locality deficit just above threshold: 0.10 vs 0.26 at $\rho = 0.25 \Rightarrow R_{o,\text{eff}} \approx 1.04$, not 1.125.



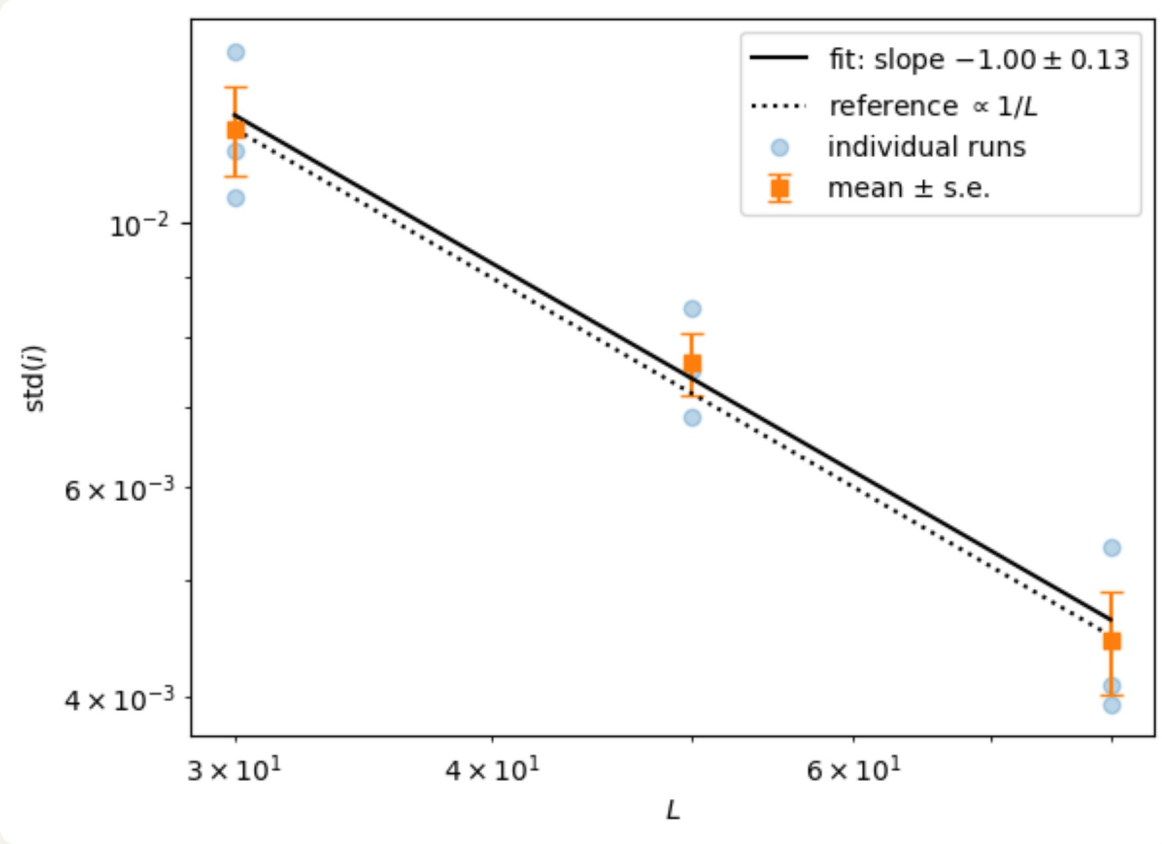
Long-term state: oscillations are noise

$b = -1.00 \pm 0.13$

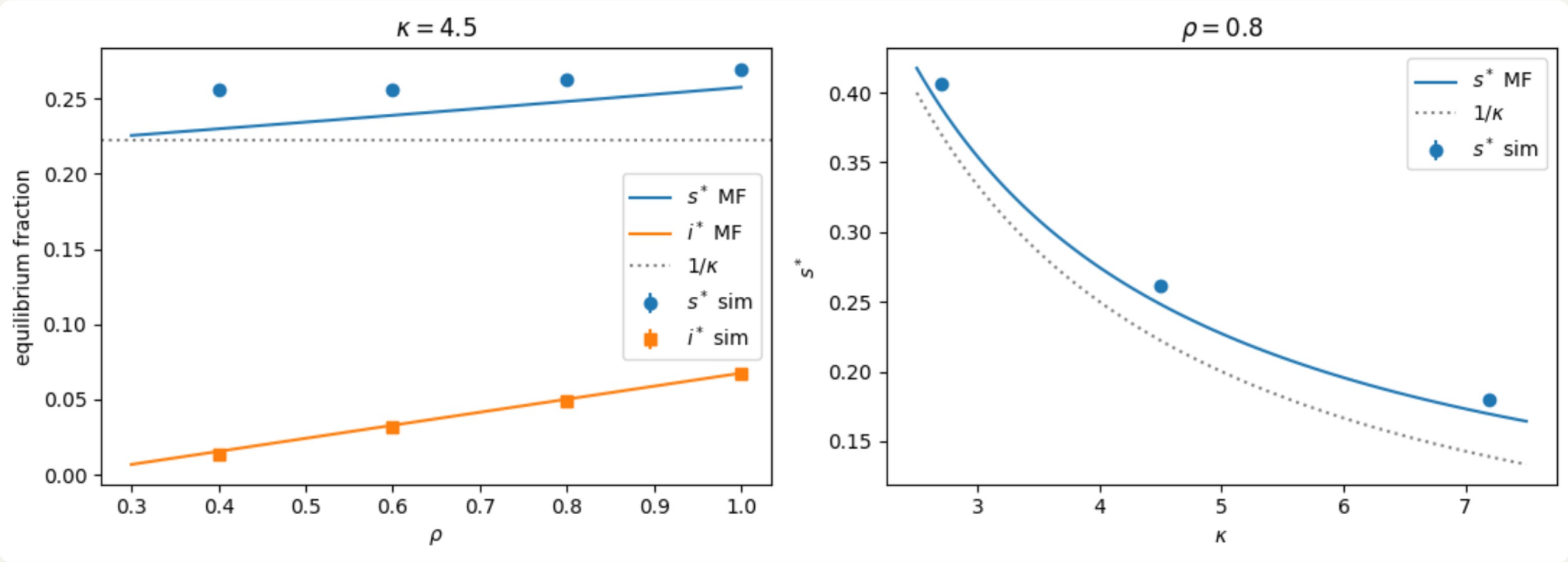
std(i) ∝ L^b · quasi-cycle −1, limit cycle 0

Period ≈ 13 generations, matching the damped mean-field mode (12.5, |λ| = 0.91).

The endemic state is a **stable fixed point**; demographic noise keeps re-exciting it. Oscillations vanish as L → ∞.



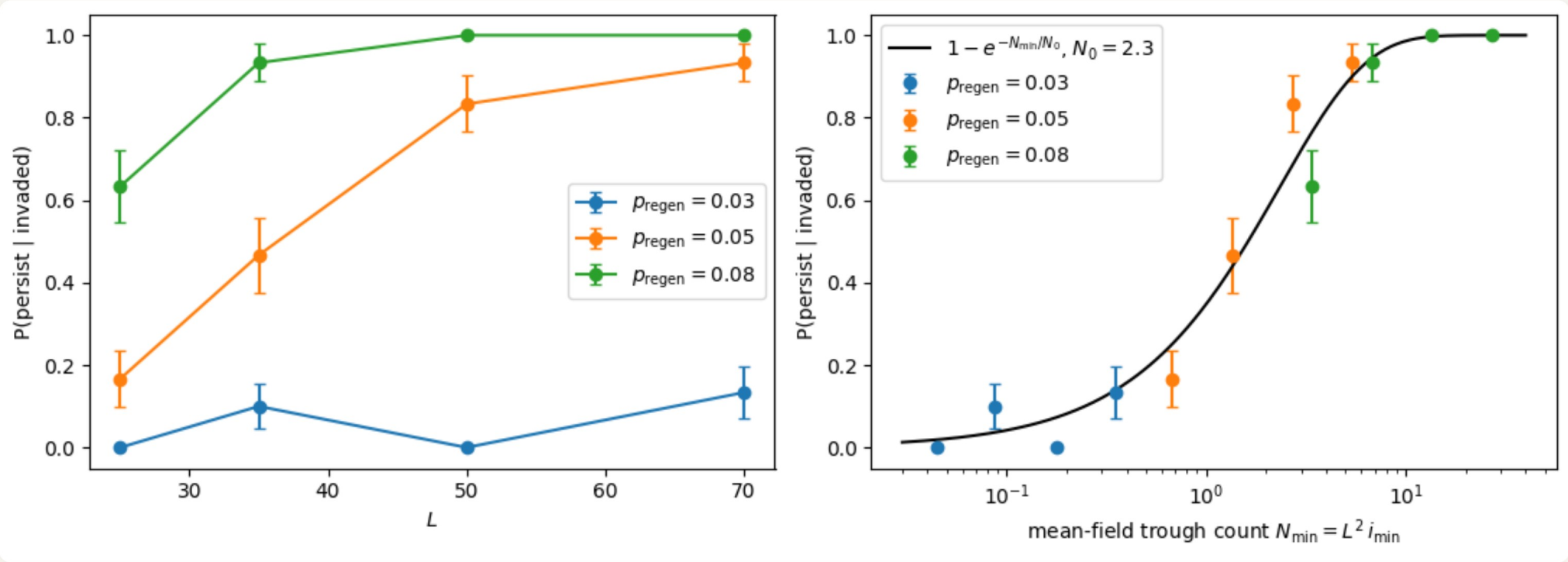
Regulation: $s^* \approx 1/\kappa$ whatever the supply



ρ × 2.5 ⇒ s* flat (0.256–0.270); i* grows 5× and matches mean-field.

Sim s* 5–11% above mean-field: locality lowers effective κ.

Extinction: one number controls persistence



Collapse onto N_min = L² i_min: P = 1 – e^(–N_min/N₀), N₀ = 2.34 ± 0.27.

Even odds at **N_min ≈ 1.6** firing traps in the first trough.

Notebook 5 in one slide

Mean-field gets right

Thresholds, equilibrium values, oscillation frequency, and the sequence invasion → overshoot → return.

Misses: demographic noise

Failed invasions above threshold, sustained quasi-cycles, first-trough extinction. All vanish as $L \to \infty$.

Misses: locality

Attempts hit spent neighbours ⇒ lower effective κ: weaker invasion, higher s^*, deeper trough.

Hypothesis to test: locality comes from **depletion**. Remove depletion and mean-field should become exact.

Neutrons as offspring, three changes

1 · No depletion

A cell is ~10^{20} nuclei: it fissions any number of times. The process becomes **linear**.

2 · Leakage

A neutron leaving the lattice is lost. Origin of a **critical size**.

3 · Delayed neutrons

A fraction β is released after a geometric delay (rate λ). Makes reactors **controllable**.

$\nu \sim P(\nu)$ from ^{235}U data: $\bar{\nu} = 2.41$, $\mathrm{Var}\,\nu = 1.22$ · $k_\infty = \rho\,\bar{\nu}\,p_f$

Sanity check: the mean grows exactly as k^t

k	sim $\langle N_9 \rangle / n_0$	k^9	deviation
0.9	0.389	0.387	0.1σ
1.0	1.001	1.000	0.04σ
1.1	2.392	2.358	0.7σ

Periodic lattice, 20 × 500 seeds. A pure Galton–Watson process: **an exact check Notebook 5 could not have**, because depletion made it nonlinear.

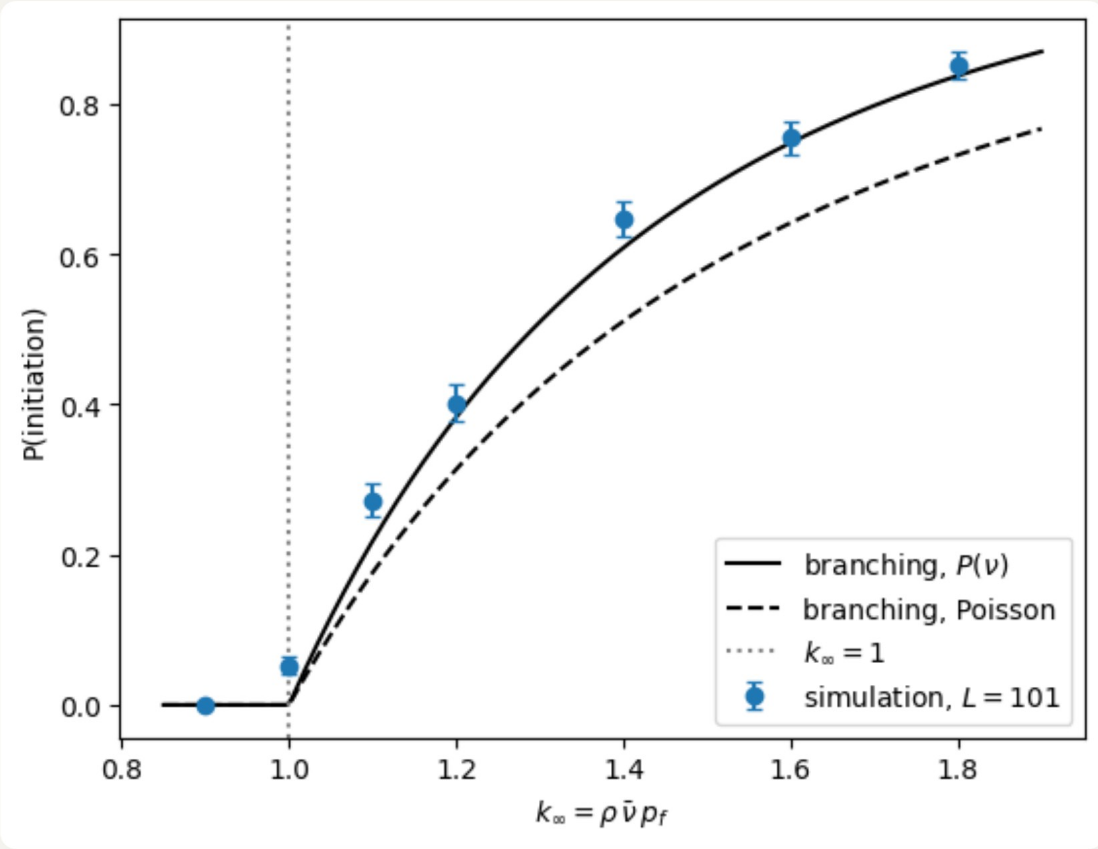
Ring sampler: 24 = 8 × 3 distinct cells at Chebyshev distance 3 — isotropic on the lattice.

Initiation: real P(ν), no locality deficit

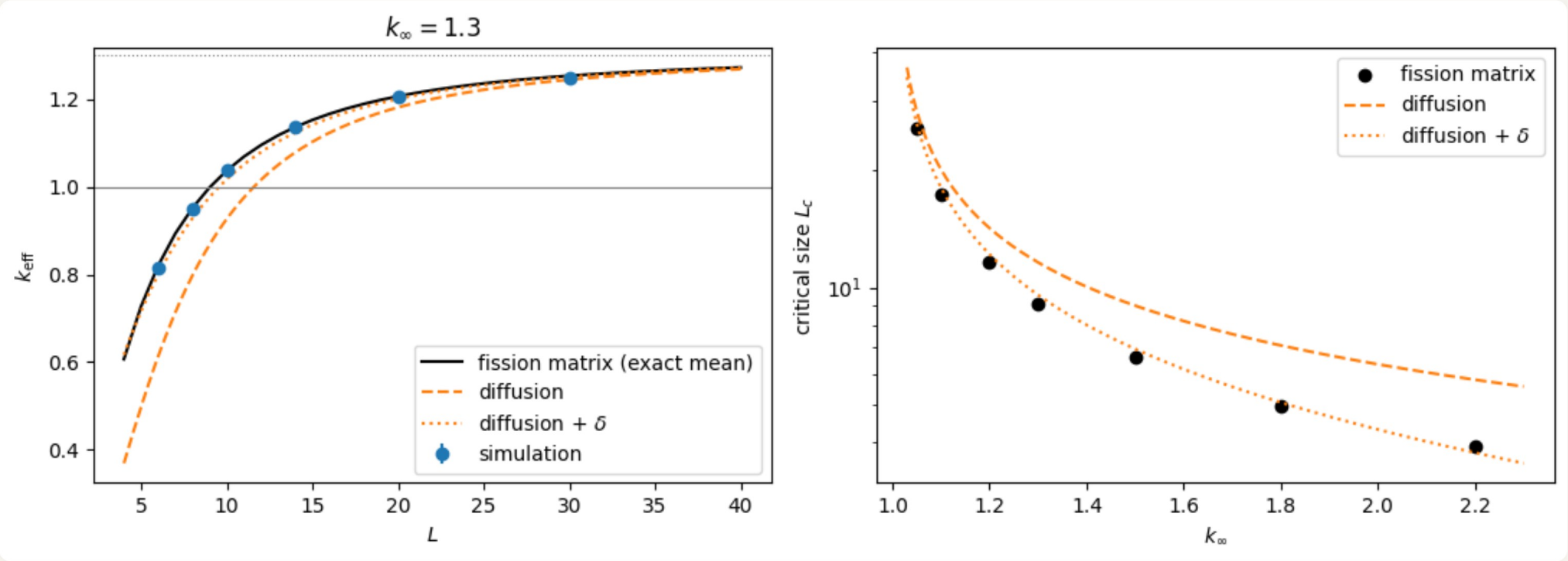
Threshold at k∞ = 1. Same transcritical kink as 5.3.

For k∞ ≥ 1.2: within **1.5σ of P(ν)**, 3.6–6.8σ above Poisson. Var ν ≈ 1.2 < $\bar{\nu}$ ⇒ fewer early deaths.

Points sit **on or above** theory, not below as in 5.3. k = 1.0, 1.1 high: finite t_c = 60 cutoff.



Critical size: leakage sets a minimum L



Sim = fission-matrix eigenvalue within ≈ 1σ for L ≤ 20. L_c ≈ 9 at k∞ = 1.3.

Diffusion overestimates L_c by ≈ 2 cells; **+ δ ≈ 0.71 M** fixes it to 0.5 cells (k∞ ≤ 1.8).

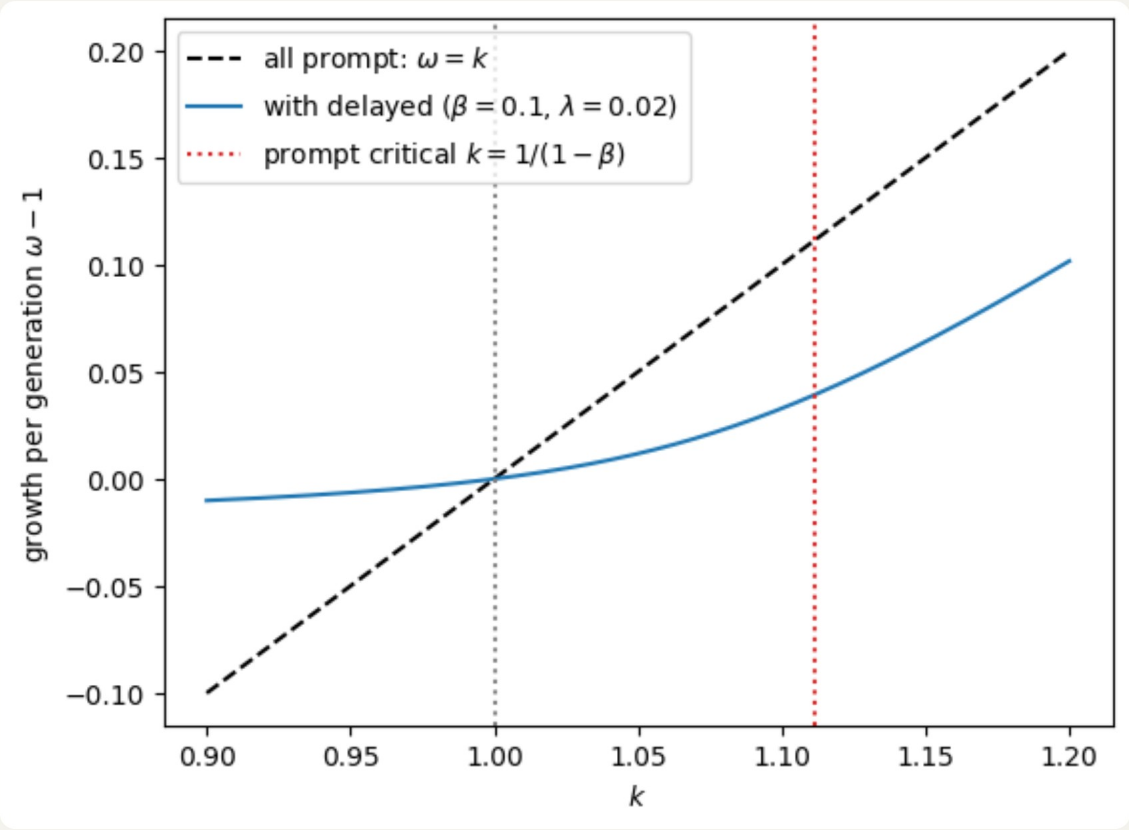
Delayed neutrons: why reactors are controllable

0.0062 vs 0.03

growth per generation at k = 1.03, with vs without delay

Critical point unchanged at k = 1. Slow growth for **1 < k < 1/(1 − β)**; fast above prompt critical.

Decay is floored at 1 − λ: the pool can only drain so fast.



Notebook 6 · 6.5 · 13 · toy β = 0.1, λ = 0.02 (mean-field map)

MAIN FINDING

Remove depletion and mean-field becomes exact in the mean.

The locality errors of Notebooks 3–5 come from a nonlinearity — attempts landing on spent sites — not from the lattice. What remains is fluctuations.

Limits and next steps

Limits

- One parameter set studied in depth (κ = 4.5, ρ = 0.8)
- Small samples: 3 runs per L (5.4), 30 per point (5.6)
- L spans only ~3×; 2D Chebyshev flights
- Single-speed neutrons, one delayed group, toy β and λ
- k_eff estimator biased at large L

Next steps

- Test locality directly: uniform-target attempts
- Restore depletion: explosive yield vs F = 1 − e^(−kF)
- Add a reflector; measure the drop in L_c
- k(N_t) feedback: self-regulation like s* ≈ 1/κ
- Start k_eff estimator from the eigenvector